HOMEWORK 3.7 · UNIT 3 · NC.8.EE.7

Inequalities & Number of Solutions

Targets: EE.7-4, EE.7-1, EE.7-7, EE.7-3, EE.7-6, EE.7-8, EE.7-9

Name _____ Date _____ Block _____

Remember: If the x's cancel: a **true** statement (5 = 5) → **infinitely many** solutions. A **false** statement (5 = 2) → **no solution**. Inequalities: solve like an equation, but **flip the sign** when you × or ÷ by a negative.
Example: $-2x > 8 \rightarrow \div(-2)$, flip: $x < -4$. Open circle for < or >. Closed circle for ≤ or ≥.

SCORE 3.0 Approaching · the basics

Targets: EE.7-4, EE.7-1, EE.7-7, EE.7-3

1 After simplifying, Omar got **7 = 7**. How many solutions? EE.7-4

One solution | No solution | Infinitely many

2 Solve. $\frac{x}{4} + 3 = 5$ REVIEW 3.5 EE.7-1

3 Which inequality matches the graph? EE.7-7



Ⓐ $x < 3$ Ⓑ $x \leq 3$ Ⓒ $x \geq 3$ Ⓓ $x > 3$

4 Solve and graph. $x - 5 < 2$ EE.7-7

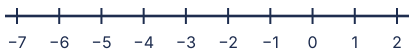


5 Solve. $5(x - 1) = 2x + 13$ REVIEW 3.4 EE.7-3

SCORE 4.0 Proficient · on target

Targets: EE.7-7, EE.7-6

6 Solve and graph. $-4x \geq 12$ EE.7-7



7 How many solutions does $5x - 2 = 5x + 3$ have? EE.7-6

(A) One
(B) None
(C) Infinitely many
(D) One: $x = 5$

8 **Leo** made a mistake. Circle it. Explain it. Fix it. EE.7-6

LEO'S WORK

$2x + 3 = 2x + 3$

$0 = 0$

So $x = 0$.

SCORE 5.0 Mastery · go deeper

Targets: EE.7-8, EE.7-9

9 Write an equation that has **infinitely many** solutions. Explain how you know. EE.7-8

My equation: _____ . I know because _____ .

10 Kai has $50. He spent $10 on parking. Tickets cost $8 each. Solve $10 + 8t \leq 50$. What does it mean? EE.7-9

t _____ . This means _____ .